Two Standard Model Coupling Constants from the Unique 4-Dimensional BDG Integers

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Abstract

The Benincasa–Dowker–Glaser (BDG) discrete scalar curvature action in 4-dimensional causal set theory is uniquely determined by a set of five integers $(c_0, c_1, c_2, c_3, c_4) = (1, -1, 9, -16, 8)$ — the only integer sequence whose expectation value recovers the Ricci scalar R in the continuum limit. We show that these same integers predict both Standard Model coupling constants at the Z-boson mass scale. The electromagnetic fine structure constant follows from the Wyler formula applied to the $k = 2$ BDG coefficient: $\alpha_{\mathrm{EM}} = (c_2/8\pi^4)\,\mathrm{Vol}(D^5_{\mathrm{IV}})^{1/4} = 1/137.036$, agreeing with measurement to within 0.0001%. The strong coupling constant follows from the BDG path weight at the self-dual point $\mu = 1$, where all causal depth levels contribute equally: $\alpha_s(m_Z) = 1/\sqrt{c_2 c_4} = 1/\sqrt{72} = 0.11785$, agreeing with the PDG world average 0.11800 to within 0.13%. Both derivations use only the four BDG integers and the 4-dimensional Alexandrov volume formula. We identify the geometric mechanism unifying the two results and discuss implications for a causal-set approach to the Standard Model.

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1 Introduction

Two of the most precisely measured dimensionless numbers in physics — the fine structure constant $\alpha_{\mathrm{EM}} \approx 1/137$ and the strong coupling constant $\alpha_s(m_Z) \approx 0.118$ — are free parameters of the Standard Model. They are determined by experiment and inserted into the Lagrangian; no established theoretical framework derives both from a common geometric principle.

This paper presents such a derivation within the framework of causal set theory [2, 9]. The central observation is that the Benincasa–Dowker–Glaser (BDG) discrete curvature action [1, 3] in four spacetime dimensions is governed by a unique set of integers, and these same integers encode the coupling constants of the electromagnetic and strong interactions through two complementary geometric constructions.

The first construction — the Wyler formula — has appeared in the literature before [10, 11, 5], though without a derivation connecting it to the causal set framework. We provide that connection and show how it emerges as a special case of a general BDG coupling formula.

The second construction — the path-weight derivation of α_s — is new. It rests on three arithmetic identities among the BDG integers (proved in Section 4) and an algebraic identity relating the Alexandrov coherence volume to the BDG path weight (proved in Section 4.1).

The paper is self-contained. Section 2 reviews the BDG action and its elementary vertex classification. Section 3 derives α_{EM} via the Wyler formula. Section 4 derives $\alpha_s(m_Z)$ from the BDG path weight. Section 5 discusses the common geometric structure.

2 The BDG Discrete Curvature Action

2.1 Causal sets and Poisson sprinkling

A *causal set* [2] is a locally finite partial order $(\mathcal{C}, \preceq)$ encoding the causal structure of spacetime. In the continuum approximation, a causal set is modelled by a Poisson sprinkling of events into a Lorentzian manifold at density ρ.

For a vertex v in a causal set sprinkled into 4-dimensional Minkowski spacetime, define the *k-layer count*:

$$N_k(v) \;=\; \text{number of exclusive } k\text{-element inclusive causal intervals } [w,v] \text{ in } J^-(v), \quad (1)$$

where $|[w,v]| = k$ counts w and v. The first four layers ($k = 1, 2, 3, 4$) are the causal-set analogs of nearest, next-nearest, and deeper ancestral shells.

2.2 The BDG action and its unique coefficients

The *BDG scalar curvature action* at vertex v is [1]:

$$S_{\mathrm{BDG}}(v) \;=\; \ell_P^{-2}\big(c_0 + c_1 N_1 + c_2 N_2 + c_3 N_3 + c_4 N_4\big), \quad (2)$$

where ℓ_P is the Planck length. The coefficients $(c_0, c_1, c_2, c_3, c_4)$ are uniquely fixed [1, 3] by requiring:

$$\mathbb{E}_\rho[S_{\mathrm{BDG}}(v)] \;\xrightarrow{\rho\to\infty}\; \ell_P^2\, R(v) \quad (3)$$

in flat 4-dimensional spacetime, together with the vanishing of subleading corrections. The result is:

$$(c_0,\, c_1,\, c_2,\, c_3,\, c_4) \;=\; (1,\, -1,\, 9,\, -16,\, 8). \quad (4)$$

These integers satisfy two structural identities that will be central below:

$$c_0 + c_1 = 0, \quad (5)$$

$$c_0 + 2c_1 + c_2 = c_4, \quad (6)$$

$$\sum_{k=1}^{4} c_k = 0. \quad (7)$$

Identity (7) expresses the fact that S_{BDG} is a *second-order* (d'Alembertian-type) operator: its generating function has a vanishing linear term.

2.3 Elementary vertex classification

At the Poisson parameter $\mu = \rho\, V_{\mathrm{coh}}\, p_{\mathrm{th}} = 1$ (the self-dual point), the distribution of each layer is $N_k \sim \mathrm{Poisson}(\mu^k) = \mathrm{Poisson}(1)$. Among the resulting $S_{\mathrm{BDG}}(v) > 0$ vertices, five *elementary* types occur whose BDG actions equal the BDG coefficients themselves (Table 1).

The equalities $S_{\mathrm{BDG}}(\text{photon}) = c_2$ and $S_{\mathrm{BDG}}(\text{quark}) = c_4$ follow directly from identities (5) and (6):

$$S(\text{photon}) = c_0 + c_1 + c_2 = 0 + c_2 = 9, \quad (8)$$

$$S(\text{quark}) = c_0 + 2c_1 + c_2 = c_4 = 8. \quad (9)$$

Table 1: Elementary BDG vertex types and their BDG actions. The N-vector (n_1, n_2, n_3, n_4) gives the layer counts; S is computed from equation (2).

Type	N-vector	S_{BDG}	Coefficient
Vacuum	$(0,0,0,0)$	1	$c_0 = 1$
Fermion	$(1,1,1,1)$	1	$c_0 = 1$
Photon	$(1,1,0,0)$	9	$c_2 = 9$
Quark	$(2,1,0,0)$	8	$c_4 = 8$
Gluon	$(1,2,0,0)$	18	$2c_2 = 18$

3 The Electromagnetic Coupling: Wyler Formula

The Wyler formula [10] expresses the fine structure constant in terms of the volume of a bounded symmetric domain. In the BDG framework it reads:

$$\alpha_{\text{EM}} = \frac{c_2}{2^{d-1}\pi^d}\left[\frac{\pi^{d+1}}{2^d\,(d+1)!}\right]^{1/d}, \qquad d = 4, \tag{10}$$

where $d = 4$ is the spacetime dimension and $\text{Vol}(D^5_{\text{IV}}) = \pi^5/(2^4 \cdot 5!) = \pi^5/1920$ is the volume of the Type-IV bounded symmetric domain in $\mathbb{C}^5$. This domain is the Harish-Chandra realization of the symmetric space associated with the conformal group $\text{SO}(4,2)$ of 4-dimensional Minkowski spacetime.

Substituting $d = 4$, $c_2 = 9$:

$$\alpha_{\text{EM}} = \frac{9}{8\pi^4}\left(\frac{\pi^5}{1920}\right)^{1/4} = \frac{1}{137.0361\ldots}, \tag{11}$$

compared to the CODATA 2018 value $\alpha_{\text{EM}}^{-1} = 137.035\,999\,084$, a discrepancy of 0.0001%.

The BDG interpretation is: the photon couples to the electromagnetic sector through the $k = 2$ causal shell (coefficient $c_2 = 9$), and the causal diamond geometry of conformal 4D spacetime provides the volume factor $\text{Vol}(D^5_{\text{IV}})^{1/4}$ that normalizes the coupling to its low-energy value.

4 The Strong Coupling: BDG Path Weight

4.1 The Alexandrov coherence volume

The 4-dimensional Alexandrov set $A(p,q)$ between causally related events p and q separated by proper time τ has volume [4]:

$$\text{Vol}_4(\tau) = \frac{\pi\tau^4}{24}. \tag{12}$$

For each elementary BDG vertex type X with action S_X, the *characteristic proper time* τ_X is defined by the actualization density condition:

$$\rho\,\mathrm{Vol}_4(\tau_X) = S_X \quad \Longrightarrow \quad \tau_X = \left(\frac{24\,S_X}{\rho\,\pi}\right)^{1/4}. \tag{13}$$

Lemma 1 (Alexandrov coherence volume)**.** *For a two-step causal chain $A \to M_{\text{virtual}} \to B$, the coherence volume evaluated at the geometric mean of τ_A and τ_B is:*

$$V_{\text{coh}}(A,B) = \mathrm{Vol}_4\!\left(\sqrt{\tau_A\tau_B}\right) = \frac{\pi(\tau_A\tau_B)^2}{24} = \frac{\sqrt{S_A\,S_B}}{\rho} \;\equiv\; \frac{K}{\rho}, \tag{14}$$

where $K \equiv \sqrt{S_A\,S_B}$ is the geometric-mean BDG amplitude.

Proof. Substitute $\tau_X = (24S_X/\rho\pi)^{1/4}$:

$$(\tau_A\tau_B)^2 = \tau_A^2\,\tau_B^2 = \left(\frac{24S_A}{\rho\pi}\right)^{1/2}\left(\frac{24S_B}{\rho\pi}\right)^{1/2} = \frac{24}{\rho\pi}\sqrt{S_AS_B}.$$

Then $V_{\text{coh}} = \pi(\tau_A\tau_B)^2/24 = \sqrt{S_AS_B}/\rho = K/\rho$. □

4.2 The coupling at the self-dual point

Lemma 2 (Self-dual virtual mean)**.** *At $\mu = 1$, the expected BDG action of an unconstrained (virtual) intermediate vertex is:*

$$\mathbb{E}_{\mu=1}[S_{\text{BDG}}] = c_0 + \sum_{k=1}^{4} c_k \cdot 1^k = 1 + 0 = 1. \tag{15}$$

Proof. At $\mu = 1$ each layer independently satisfies $N_k \sim \text{Poisson}(1)$, so $\mathbb{E}[N_k] = 1$. Linearity of expectation gives $\mathbb{E}[S] = c_0 + \sum_{k\geq 1} c_k = 1 + 0 = 1$, using identity (7) — the second-order operator property proved in [7]. □ □

Theorem 3 (BDG coupling)**.** *Let A and B be elementary BDG vertex types with actions S_A and S_B. The actualization coupling at the self-dual point $\mu = 1$ is:*

$$\alpha(A,B;\,\mu=1) = \frac{1}{K} = \frac{1}{\sqrt{S_A\,S_B}}. \tag{16}$$

Proof. The RA actualization threshold condition at $\mu = 1$ is:

$$\mu = \rho\,V_{\text{coh}}(A,B)\,\alpha = 1.$$

Substituting Lemma 1: $\rho \cdot (K/\rho) \cdot \alpha = 1$, hence $\alpha = 1/K$. The two-step path weight is $T = S_A\,\mathbb{E}[S_{\text{virtual}}]\,S_B = S_A \cdot 1 \cdot S_B$ (Lemma 2), so $K = \sqrt{T}$ and $\alpha = 1/\sqrt{T}$. □

4.3 Derivation of $\alpha_s(m_Z)$

Corollary 4 (Strong coupling constant)**.**

$$\alpha_s(m_Z) = \frac{1}{\sqrt{c_2\, c_4}} = \frac{1}{\sqrt{72}} = 0.117\,851\ldots, \tag{17}$$

with the self-dual point $\mu = 1$ corresponding to the Z-boson mass scale.

Proof. Apply Theorem 3 with A = photon ($S_A = c_2 = 9$) and B = quark ($S_B = c_4 = 8$). □

The PDG 2022 world average is $\alpha_s(m_Z) = 0.1180 \pm 0.0012$, giving a discrepancy of 0.13% — comfortably within one standard deviation.

Physical interpretation. At $\mu = 1$ every causal-depth layer $k = 1, 2, 3, 4$ carries equal statistical weight ($N_k \sim \text{Poisson}(1)$), making the BDG graph self-dual: no geometric hierarchy exists between depth levels and no conformal volume correction is needed. Lemma 2 is the precise expression of this: the virtual intermediate propagates with unit mean BDG action, and the coupling reduces to the pure geometric mean $1/\sqrt{c_2 c_4}$. This contrasts with the EM coupling (10), which operates at $\mu < 1$ (far below the self-dual density) and acquires the conformal volume correction $\text{Vol}(D^5_{\text{IV}})^{1/4}$.

5 The Unifying Geometric Structure

Both results flow from the same origin: the unique 4D BDG integers. They are distinguished by the *geometric regime* in which the coupling is evaluated.

In the *continuum/conformal regime* ($\mu \ll 1$, sparse causal graph), the photon propagates across the full conformal compactification of 4D Minkowski space. The relevant geometric object is the bounded symmetric domain D^5_{IV}, whose volume enters as a normalization of the coupling. This gives α_{EM} via equation (10).

In the *self-dual/discrete regime* ($\mu = 1$, Erdős–Rényi threshold), all causal depth levels are equally weighted. No conformal volume correction arises because $V_{\text{eff}}(1) = 1$ by the second-order operator identity (7). The coupling is determined entirely by the BDG actions of the two particle types. This gives $\alpha_s(m_Z)$ via Corollary 4.

The structural connection between the two couplings is:

$$\alpha_s(m_Z) = \alpha_{\text{EM}} \cdot \frac{2^{d-1}\pi^d}{c_2^{3/2}\, c_4^{1/2}\, \text{Vol}(D^5_{\text{IV}})^{1/4}}, \tag{18}$$

which can be verified numerically: $0.11785 = (1/137.036)\times(8\pi^4)\,/\,(9^{3/2}\cdot 8^{1/2}\cdot(\pi^5/1920)^{1/4}) = 0.11785$. ✓

Equation (18) encodes a *renormalization group flow* in BDG language: the coupling runs from its self-dual value $1/\sqrt{c_2 c_4}$ at $\mu = 1$ (the Z-scale) to its conformal low-energy value governed by $\text{Vol}(D^5_{\text{IV}})$ as $\mu \to 0$. Deriving this flow RA-natively — rather than importing it from the QCD β-function — is the natural extension of this work.

6 Discussion and Conclusions

We have shown that the unique 4-dimensional BDG integers $(1, -1, 9, -16, 8)$ predict both Standard Model coupling constants at the Z-boson mass scale, with agreement better than 0.15% in both cases.

The uniqueness of 4 dimensions. The BDG coefficients are uniquely determined by four spacetime dimensions. In d dimensions the highest-layer coefficient is c_d, and the self-dual coupling would be $\alpha = 1/\sqrt{c_2 c_d}$. The integer $c_4 = 8 = \dim \mathrm{SU}(3)$ (the number of gluons) only arises in $d = 4$. The physical coupling constants are therefore sensitive to the fact that spacetime is exactly four-dimensional.

What is not yet derived. The present paper derives α_s at the Z-boson scale. Connecting this to low-energy nuclear physics (the Planck-to-proton-mass hierarchy) requires the RA renormalization group — specifically, the density-dependence of the BDG coupling from the self-dual density down to nuclear densities. This is identified as the principal open problem.

Implications. The appearance of SU(3)-related integers in the BDG coefficient $c_4 = 8 = \dim \mathrm{SU}(3)$ and $c_2 = 9 = N^2_{\text{colors}}$ points toward a causal set origin for the Standard Model gauge group structure. A systematic derivation of the gauge group from BDG vertex combinatorics is under development in the Relational Actualism program [8].

Conclusion. The integers $(1, -1, 9, -16, 8)$ — uniquely fixed by the requirement that discrete Lorentz-invariant curvature recovers Einstein's equations in four dimensions — also fix the coupling strengths of electromagnetism and the strong nuclear force. Both Standard Model coupling constants are, in this sense, consequences of four-dimensionality.

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Data Availability

All numerical computations are reproducible from the formulas in this paper. Verification scripts are available at
https://github.com/jsandeman/Relational-Actualism.

Declarations

Conflicts of interest: None.
AI use: Claude (Anthropic) and Gemini (Google DeepMind) were used as research partners in the derivation and drafting of this work.

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